

Adam Driscoll

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Software engineer with experience in robotics, georegistration, localization, navigation, mapping, and 3D reconstruction.

Work Experience

Software Engineer

Palantir | Python

Oct 2021–Jan 2026

- Designed and built Unscented Kalman Filter that incorporates georegistration to estimate in-flight aerial vehicle global position in gps-denied settings
- Integrated the estimation algorithm into a simulation environment by utilizing Docker containers and the MAVlink protocol. Wrote a script to automatically generate and launch a flightpath in this environment
- Developed optical flow algorithm to enable real-time tracking of frames in full motion videos by updating frame coordinates in-between executions of georegistration algorithm
- Adapted internal python module capabilities to integrate georegistration tools into Palantir's Sensor Inference Platform, allowing it to output corrected geo-coordinates
- Implemented subpixel registration algorithm to improve the accuracy of our satellite georegistration process

Computer Vision Engineer II

Hivemapper | C++, Python

2019–2021

- Designed, built, and maintained Open Telemetry Kit, an open source python tool to automatically extract drone camera and flight data to feed into the Hivemapper platform to improve 3D modeling and georegistration
- Developed algorithm to determine confidence in point cloud vertices and remove low confidence points, improving final point cloud and mesh quality
- Enhanced approach to detecting and masking video overlays to video and extended this algorithm to apply during mesh texturing
- Investigated and validated various SLAM, VO, and depth estimation algorithms to improve accuracy and quality of maps generated from dashcam videos

Software Developer, Intern

Discovery Robotics | ROS, Gazebo

2018

- Built simulation environment in Gazebo to enable rapid iteration on software algorithms
- Modeled flagship robot in simulation including accurate physics, motion model & sensor representation

Operational Stability Engineer

Amazon Robotics | MySQL

2013-2017

- Developed over 20 automation tools to replace manual task execution & reduce system failures
- Analyzed over 200 complex software issues & identified their root causes
- Created MySQL queries to collect data from 29 client facilities & presented this data in a UI using internally developed tools, allowing maintenance teams to efficiently analyze warehouse status

Education

- **M.Sc. Robotic Systems Development**, Carnegie Mellon University, Pittsburgh, PA 2018
- **B.Sc. Computer Science**, Worcester Polytechnic Institute, Worcester, MA 2012

Projects

Groundsbot, Capstone

CMU, groundsbot.com | C++, ROS

- Designed, built, and programmed an autonomous field robot capable of mowing the rough grass at a golf course
- Created robust perception & localization subsystems by fusing data from Lidar, RTK GPS, IMU & encoders
- Created GPS waypoint following & control algorithms used in the navigation subsystem of GroundsBot
- Achieved 98.6% coverage of input area, avoidance of 4/5 static obstacles & detection of 22/25 dynamic obstacles